BLACK PANTHER

BPX USER MANUAL



1. Safety Notes

1.1 Environmental Requirements

(1) Choosing a Suitable Environment

Because this product generates substantial kinetic energy when running and braking, its braking distance may exceed 1 m. Always stay away from crowds and use the product in an open, spacious area, whether indoors or outdoors.

(2) Be Aware of the Surroundings

This product is mainly made of metal. When using it, pay attention to the surroundings to avoid unnecessary injury to yourself, nearby objects, people, or small animals.

(3) Power Adapter Precautions

The China-specification power adapter is designed for use at altitudes below 2000 m in non-tropical regions. Adapters of other specifications are not subject to this restriction.

(4) Severe Weather and Disaster Conditions

Do not use the product in severe weather such as strong wind, snow, rain, thunderstorms, or fog. Do not use the product during disasters such as fire, explosion, lightning strike, storms, tornadoes, heavy rain, floods, earthquakes, or sandstorms.

(5) Unsuitable Surfaces

Do not use the product on muddy, loose, slippery, icy, or puddle-covered surfaces.

(6) Water Resistance and Water Exposure

This product is not waterproof. Do not immerse the body in water, and avoid cleaning it with strong water streams or high-pressure hoses to prevent the risk of fire or explosion.

If the product is exposed to rain or accidentally falls into water while powered on, disconnect the power immediately to avoid continuous energization. Continued use may cause permanent damage to the product and may even pose a safety hazard. Please operate with caution and take appropriate protective measures as soon as possible. If the product is exposed to rain or falls into water while powered off, remove the power source immediately and avoid attempting to restart or charge it. This prevents damage to the internal circuits and reduces the risk caused by accidental contact with water.

(7) Electromagnetic Interference

Do not use the product in environments with frequent wireless communication activity. Also, when using it in areas with poor communication signals or strong electromagnetic interference, be aware of possible effects on the sensors and communication signals.

Building walls, rocks, and trees may interfere with the sensors and communication signals. Use the product in an open, spacious area.

Using the product near electromagnetic interference sources (such as Wi-Fi hotspots, routers, Bluetooth devices, high-voltage power lines, high-voltage substations, mobile phone base stations, and TV broadcast towers) may affect control performance.

(8) Temperature and Heat Sources

Do not place the product in fire or heaters, and avoid using it in high-temperature environments to prevent battery overheating, fire, or functional failure.

1.2 Usage Precautions

(1) Joint Inspection

Check that all four legs are complete and that each joint moves freely without obstruction from foreign objects.

(2) Battery Level

Check the battery level to make sure it is sufficient, to avoid sudden collapses or falls.

(3) Proper Placement and Ground Conditions

When placing the product, make sure the ground is level so that the robot dog lies flat on the ground. Avoid placing it on hard, sharp, or overly uneven surfaces.

(4) Safe Use

Keep the product within your line of sight during use and maintain a sufficient safe distance, away from obstacles and anything it might collide with.

Do not let two or more people control the robot dog at the same time.

While the product is powered on, do not push, pull, or carry the robot dog by its legs to avoid pinching your hands.

(5) Speed, Turning, and Complex Terrain

Turning quickly can easily cause loss of control, especially on steep slopes or uneven or unknown terrain. Avoid risky maneuvers and reduce speed. Although the product has good terrain adaptability, always check whether the terrain is suitable.

1.3 Battery and Charging

(1) Battery Usage Restrictions

Do not disassemble the battery or draw out the battery terminals for use in any way.

(2) Adapter and Cleaning

Do not clean the power adapter with alcohol or other flammable agents to avoid potential safety risks.

(3) Handling Battery Leakage

If the battery leaks and electrolyte gets into your eyes, do not rub them. Rinse your eyes immediately with clean water and then seek medical treatment.

(4) Handling Abnormal Conditions

If the battery emits an unusual odor, smoke, deforms, or shows other abnormal signs during charging, unplug the power adapter immediately and inspect it.

(5) Safe Temperature

The power adapter may become hot during charging. Avoid prolonged contact to prevent burns.

(6) Power Adapter

Do not use a damaged power adapter. Only use the official power adapter. Non-official power adapters may cause the lithium battery to leak, overheat, catch fire, or explode.

(7) Charging Time

Do not charge the battery for more than 12 hours; overcharging may shorten battery life. Once the power adapter indicator turns green, you may unplug the charger.

1.4 Safe Use

(1) Using the product outdoors at night is more dangerous than during the day, as others may hardly notice it, which can be hazardous. It is recommended not to use the product at dawn, dusk, or night.

(2) Stay alert and safety-conscious while using the product, and always watch for passing vehicles and pedestrians. Be prepared at all times for the following situations: 1. someone suddenly approaching; 2. children or pets playing nearby; 3. potholes, drain covers, railway tracks, expansion joints, road construction, and surrounding obstacles. These may lead to accidents.

(3) Before use, make sure you fully understand this product. Do not let people unfamiliar with its operation use it.

(4) As wires, small parts, and movable mechanical joints may pose a danger to children, keep children at a safe distance of more than 1 m from the product while the robot dog is moving.

(5) Do not disassemble or modify the electronic control box of this product. Use only official parts; non-official parts may be incompatible with the software, causing motion control failure and making the product unusable.

(6) Do not use the product to ram into buildings, crowds, or animals, and do not use it to carry, drag, or launch any dangerous objects.

(7) Do not operate the product when under the influence of alcohol, drugs, or anesthesia, or when experiencing dizziness, fatigue, nausea, or any other poor physical or mental condition.

(8) If the product is involved in an accident (e.g., collision, fall) or operates abnormally (e.g., unstable standing or loss of control), stop using it immediately. Do not continue using the product until it has been repaired by after-sales service personnel.

(9) Do not use the product on public roads so as not to affect other pedestrians or vehicles.

(10) Do not exceed a payload of 5 kg.

(11) Do not use the product in crowded places.

1.5 Maintenance and Transportation

(1) Visual Inspection

Check that the product is intact and undamaged, paying special attention to any impact damage. If in doubt, contact after-sales service personnel.

(2) Regular Maintenance and Cleaning

Before cleaning, make sure the product is powered off and the power adapter is disconnected.

Regularly clean dust and other debris from the product surface and components.

(3) Part Replacement

When replacing parts, always use officially designated parts to avoid damaging the product due to incompatibility.

(4) Storage Environment

Store the power adapter and the product in a dry, cool environment, away from direct sunlight, fire sources, and strong electromagnetic fields. The recommended storage temperature is 20°C to 25°C.

Do not store the product in environments above 60°C or below -10°C. For long-term storage, a temperature of 20°C-25°C and a relative humidity of 0-60% RH are recommended, with the total battery level ideally kept below 50%.

(5) Battery Precautions

Prolonged inactivity may affect battery performance. Avoid deep discharge to keep the battery in good condition.

Check the battery level regularly.

If the battery level drops below 20%, charge it to 40%-60% before storage to avoid battery damage.

(6) Handle with Care

After powering on, avoid lifting or carrying the product. Always power it off before carrying it.

Handle the product gently when carrying it to avoid mechanical shock that may damage components.

1.6 Laws and Regulations

(1) Do not use the product for any illegal or improper activity (including but not limited to espionage, military operations, and illegal investigations).

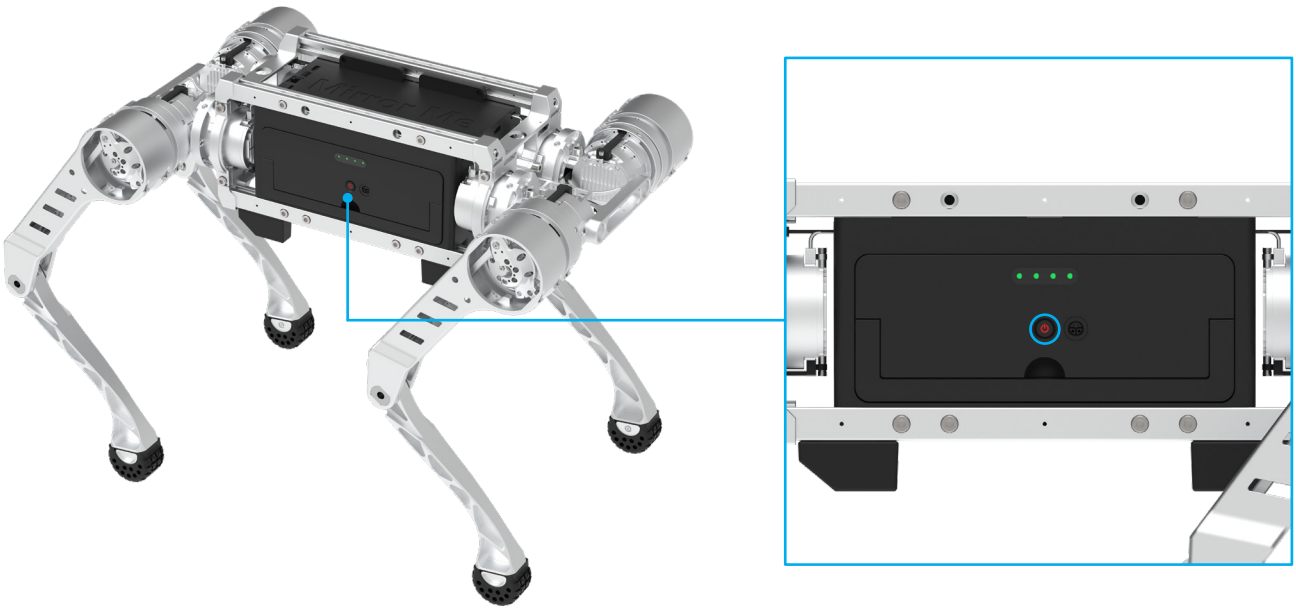
(2) Do not use the product to infringe on others property rights.

(3) Do not use the product to invade others privacy. Always respect others privacy rights when taking photos or recordings with the product. Do not use the product for any surveillance activity, including but not limited to surveillance of individuals, groups, events, performances, exhibitions, or buildings.

(4) Do not operate the product near or inside sensitive buildings or facilities, including but not limited to border lines, power plants, substations, prisons, main traffic routes, railway and highway transport hubs, government buildings, and military bases.

(5) Users are obligated to be aware of and comply with the laws and regulations of their region. If local laws and regulations impose other requirements on the use of electronic products such as robots and cameras, use the product in accordance with those requirements.

2. Power On / Off



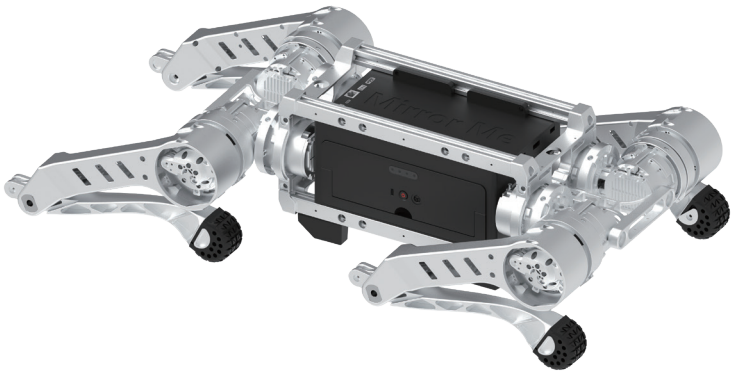
Power button illustration

2.1 Power On

Prerequisite: Make sure all four legs of the robot dog are in the **standard lying-down posture.**

Powering on: Locate the power button on the side of the body. **Press the power button briefly once, then press and hold it for 3 seconds** to power on. At the same time, you will see the power indicator light up and flash sequentially, showing the current battery level.

First power-on: After assembling the robot dog and powering it on for the first time, use BAOBOT APP > **Fault Diagnosis** to check whether the assembly is correct, then use APP > Remote Control > **Reset Joints** to zero the joints. See Section 3, APP Remote Control, for details. Otherwise, the robot dog will not move properly.



Standard lying-down posture

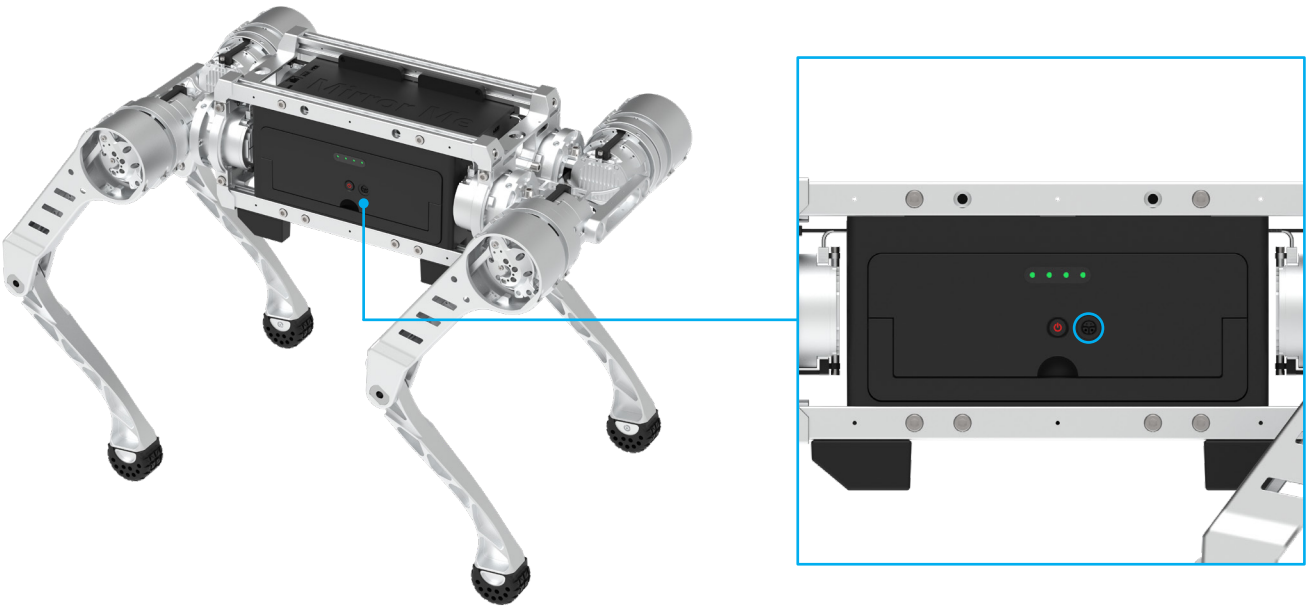
2.2 Power Off

Prerequisite: Make sure the robot dog has completely stopped moving and is in the **standard lying-down posture.**

Powering off: Locate the power button on the side of the body. **Press the power button briefly once, then press and hold it for 3 seconds.** Once the power indicator goes out, the robot dog will power off automatically.

2.3 Charging

The robot dog is charged by direct connection to the charger. Direct charging is the most efficient and easiest charging method. No matter the environment, as long as you use the charger kit provided by Mirror Me Technology, you can quickly recharge the robot dog to keep it running efficiently.

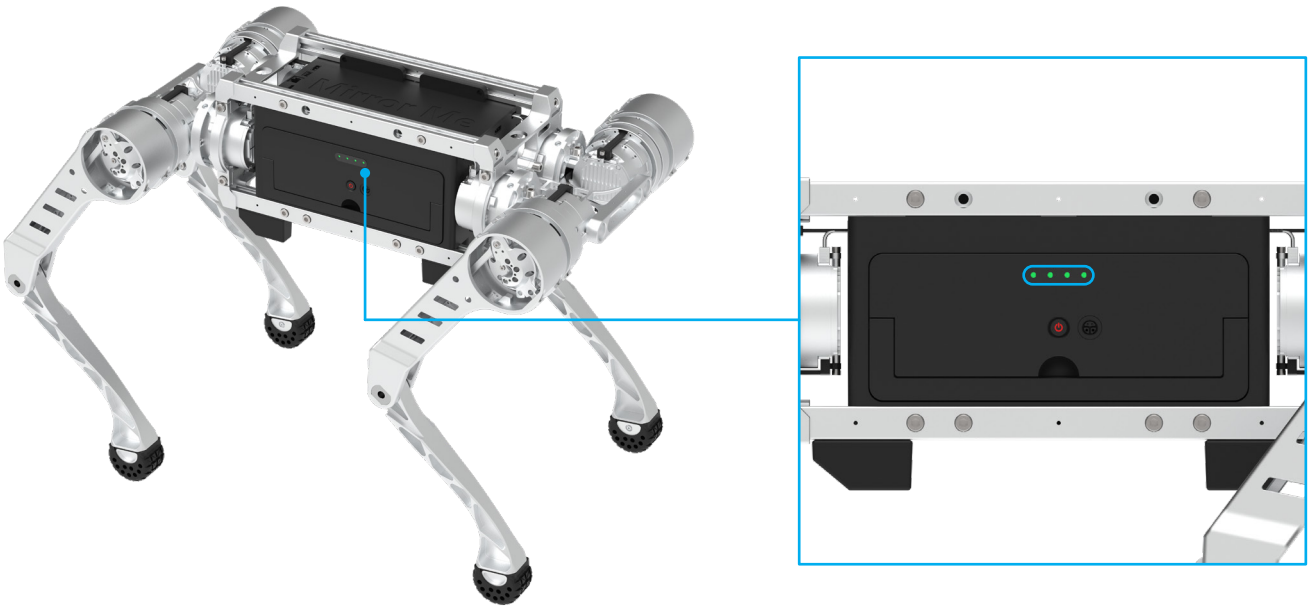


Power port

2.4 Checking the Battery Level

Via the power indicator: When the robot dog is powered on, the four LED lights on the device show the current remaining battery level in an intuitive way.

Via the APP: When the robot dog is powered on, you can connect to it with the APP and check the remaining battery level on the Device page.



Power port

Battery Level Display	**4 green LEDs solid on**	Remaining battery >75%
	3 green LEDs solid on	50% < Remaining battery ≤75%
	2 green LEDs solid on	25% < Remaining battery ≤50%
	1 green LED solid on	0% < Remaining battery ≤25%
Charging Status	**4 green LEDs in running light sequence**	Charging in progress
	All 4 green LEDs off	Fully charged
Power On / Power Off	**4 green LEDs light up sequentially**	Power ON
	All 4 green LEDs turn off simultaneously	Power OFF

3. APP Remote Control

3.1 Connecting via the APP

1. For first-time use, scan the QR code with your phone or download the BAOBOT APP from the app store.

- APP download

  iOS system requirements: iOS 12.0 or later

  Android system requirements: Android 7.0 or later

iOS download QR code

Android download QR code

2. Open the APP and log in with one-tap phone-number login or the official account and password.

3. Tap Device, then the Add button in the upper-right corner, and scan the SN code to add the device.

a. SN code location: on the side of the robot dog control box (near the right front leg)

b. Entering the SN code: scan the SN code / upload a photo of the SN code taken with your phone / enter the SN code manually

4. In APP > Device, tap Device List in the upper-left corner and select the device you want.

5. In Phone > Settings > Wi-Fi, connect to the robot dog hotspot Wi-Fi.

a. Wi-Fi name: BPX_XXXXXX

b. Wi-Fi password: 12345678

6. Tap the Connect button in APP > Device to connect to the robot dog, until the message Connected appears on the APP page.

3.2 Remote Control via the APP

Entry: APP > Device > Remote Control

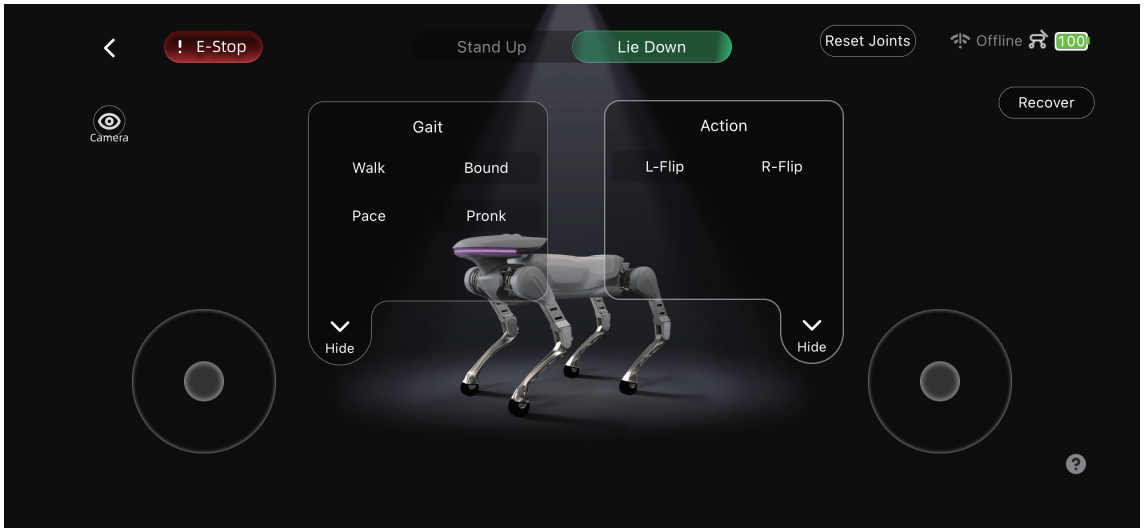
Gaits: Walk, Bound, Pace, Pronk

Actions: L-Flip, R-Flip

Reset Joints: Adjust the robot dog legs to the **standard lying-down posture**, then tap Reset Joints to zero the joint motors.

Stand up after falling: If the robot dog falls, use this button to restore it to the **standing posture**.

Soft emergency stop: If the robot dog becomes uncontrollable, the operator should immediately press the emergency stop switch. This immediately triggers the protection mechanism and makes the robot dog stop all movements.

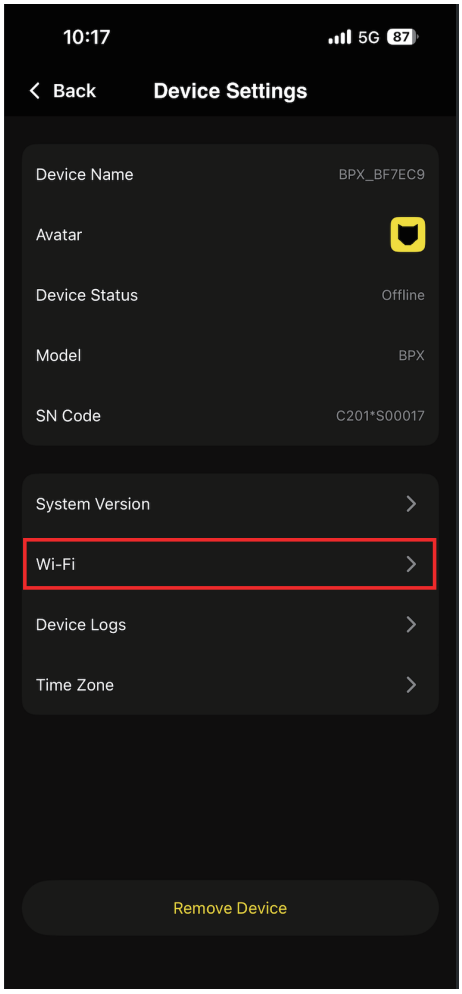


APP > Device > Remote Control

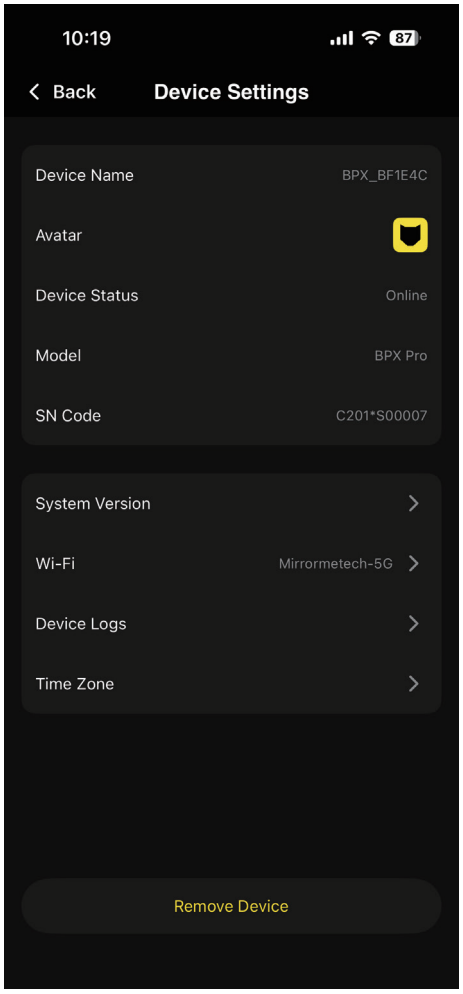
4. Network Configuration

Path: Connect the device in the APP, then APP > Device > Device Settings > Wi-Fi Network

Operation: Select the target Wi-Fi, enter the password, and once configuration succeeds, you will see the configured network name under Wi-Fi Network.



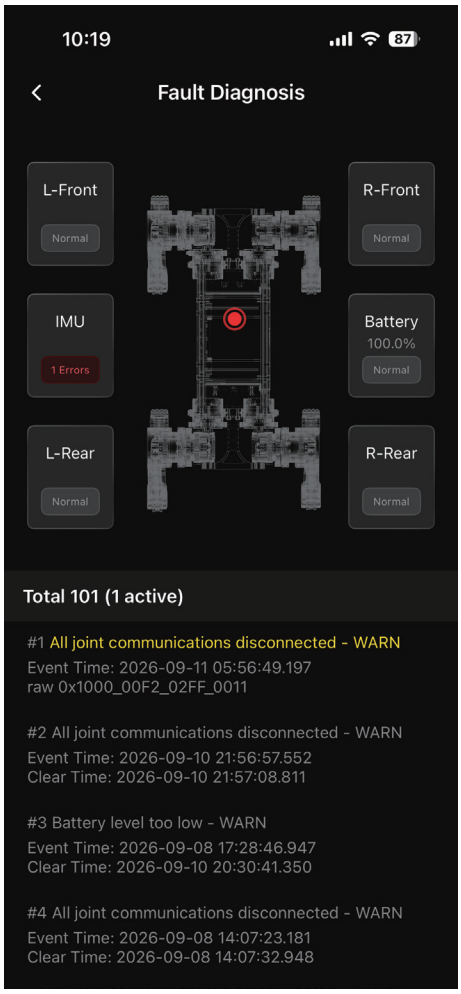
Network configuration entry



Network configuration success

5. Fault Diagnosis

Path: Connect the device in the APP, then check the device status in APP > Device > Fault Diagnosis and resolve any faults following the guidance.



Fault Diagnosis

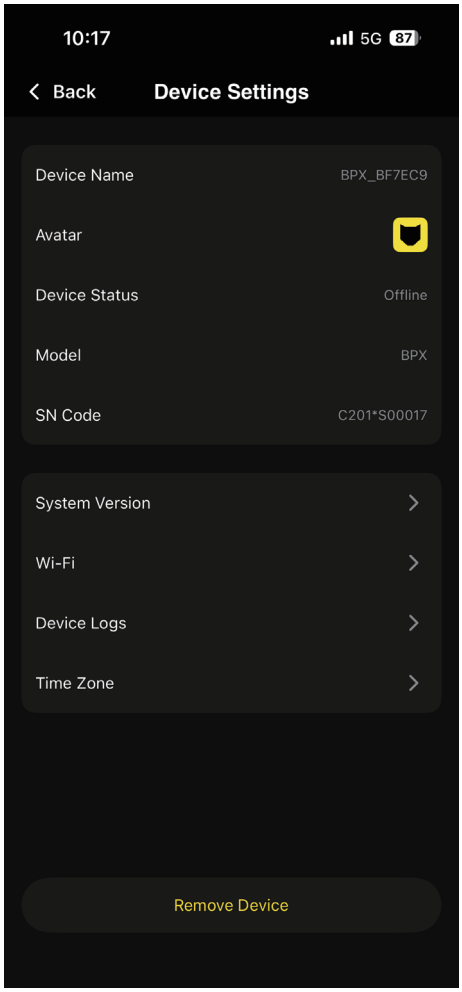
Common faults and solutions are listed below:

Fault Diagnosis	Solution
M-XX joint communication lost	Check whether the motor cable of this joint motor or its upstream motor is loose or disconnected
M-XX joint position out of range	Lay the robot dog down, perform a joint reset, then try again
Mismatched knee joint models	Standard and PRO knee joint motors are mixed. Make sure all four knee joints use the same motor model.
M-XX joint installed in the wrong position	Following the guidance, remove the incorrectly installed joint and reinstall the correct one.
M-XX motor over-temperature	Lay the robot dog down, let it rest for a while, then try again
Low battery level	Lay the robot dog down and charge the battery

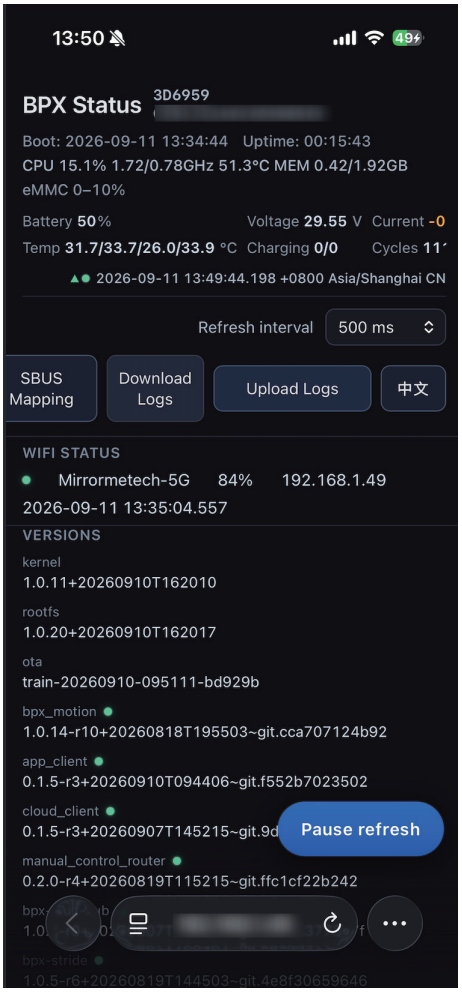
6. Device Logs

Path: Connect the device in the APP, then APP > Device > Device Settings > Device Logs to upload or download logs.

Log download: If the robot dog encounters an unknown fault, follow the official technical support guidance to package and download the logs, then send the downloaded logs to technical support for review.



Device log entry

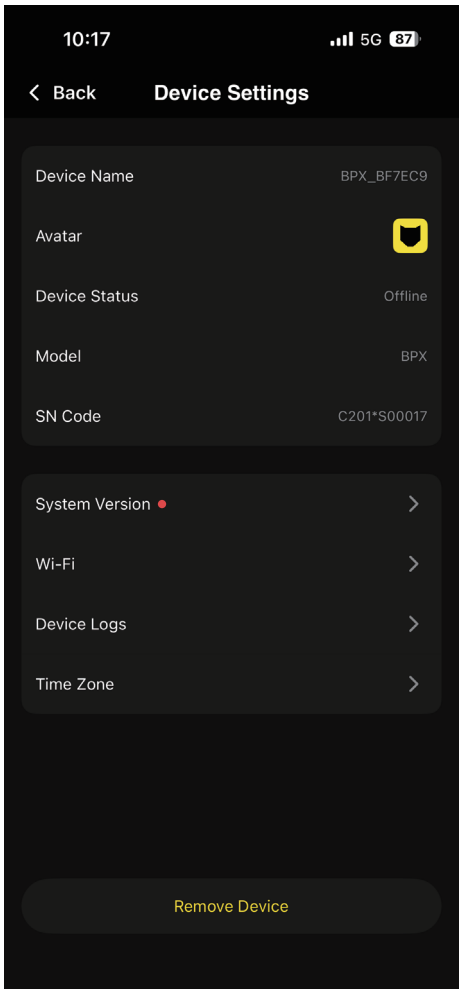


Log upload / log download page

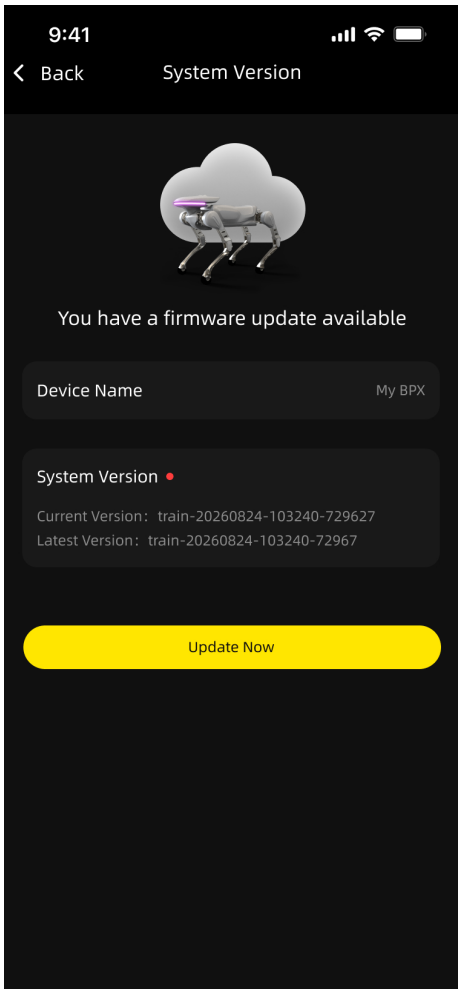
7. Firmware Upgrade (OTA)

Path: Connect the device in the APP, then APP > Device > Device Settings > System Version, and update the firmware following the guidance.

Note: Before updating, make sure the robot dog has been connected to the network (see Section 4, Network Configuration) and that its legs are in the **standard lying-down posture**. Do not operate the robot dog during the update.

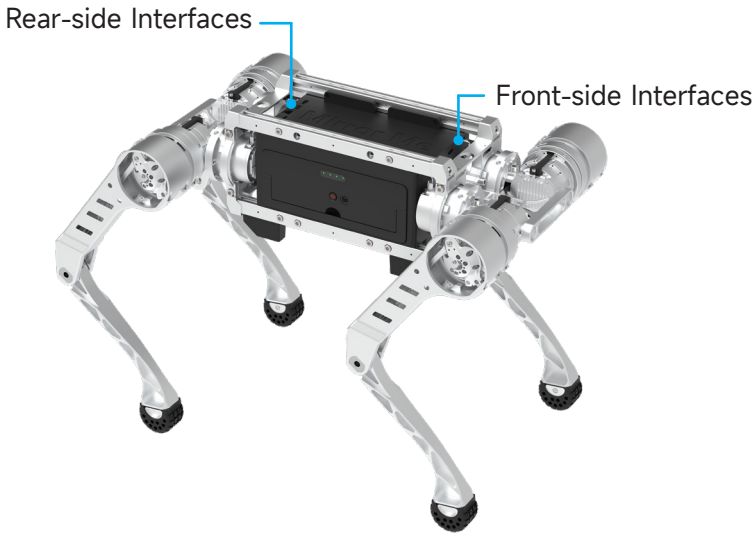


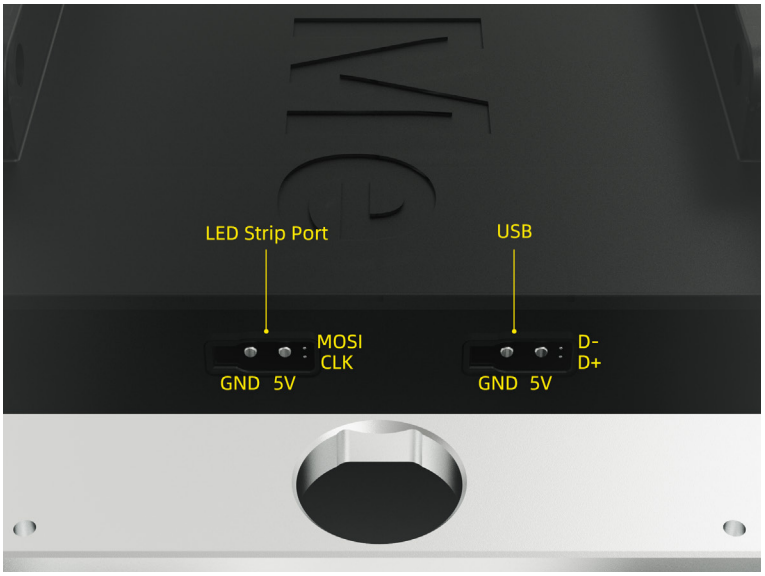
Firmware upgrade entry



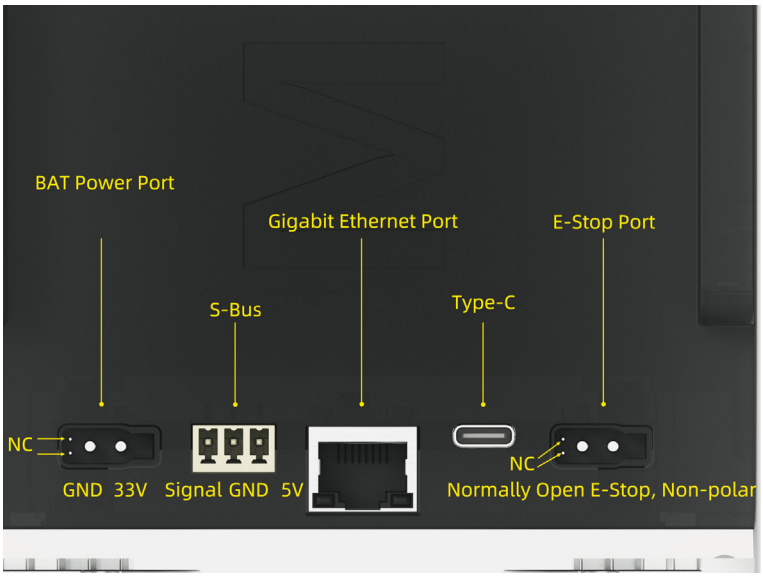
Firmware upgrade page

8. Interface Description





Front-side Interfaces



Rear-side Interfaces

9. FAQ

Q1 What should I do if the robot dog shows no response after pressing the power button?

(1) Confirm the battery is properly installed: Push the battery fully home to prevent failure to power-on or hotspot search issues caused by poor contact.

(2) Verify correct power-on operation: Locate the side power button on the unit. Short-press the power button once, then long-press it for 3 seconds. The power indicator lights will light up sequentially.

(3) If there is still no response after the above checks, contact after-sales service personnel.

Q2 Can I lift or move the robot dog after power-on?

No. Avoid lifting or moving the product once powered on. Do not pull or push the robot dog's limbs to prevent hand pinching injuries or abnormal movements. Always power off the unit before moving it. Handle with care during transportation to avoid component damage from mechanical shock.

Q3 What can I do if my phone cannot find the robot dog's hotspot Wi-Fi?

(1) Ensure the robot dog has successfully booted up. Wait about one minute after power-on for the hotspot to become discoverable.

(2) Check for poor battery contact; make sure the battery is pushed fully into place.

(3) Restart the robot dog and search for the hotspot Wi-Fi again. The robot dog hotspot name is BPX_XXXXXX, and the default password is 12345678.

Q4 Can two or more people control the robot dog simultaneously?

No. Do not allow two or more persons to operate the robot dog at the same time. Conflicting commands may lead to loss of control or accidents.

Q5 What does alternating red-green flashing on the charger indicator mean?

It indicates abnormal battery temperature. Stop charging immediately. Resume charging approximately 10 minutes later once the battery has cooled down.

Q6 What precautions apply to charging?

(1) Only use the official power adapter / charger kit. Do not use damaged or non-official adapters. Improper adapters may cause lithium-ion battery leakage, overheating, fire or explosion.

(2) Charging duration must not exceed 12 hours. Over-charging may shorten battery service life. Unplug the power cord once the adapter indicator turns green.

(3) The power adapter can become hot during charging. Avoid prolonged skin contact to prevent burns.

(4) If the battery emits strange odors, smokes or deforms while charging, unplug the adapter right away and inspect the device.

(5) Do not clean the power adapter with alcohol or other flammable agents.

Q7 How should batteries be managed for long-term storage?

Prolonged idle storage can degrade battery performance. Avoid deep discharge to preserve battery health.

(1) When charge level drops below 20 %, charge the battery to 40 %-60 % before storage.

(2) For long-term storage, maintain ambient temperature between 20 °C-25 °C and relative humidity from 0-60 %RH. Keep battery capacity below 50 % and inspect battery charge periodically.

(3) Do not store the product in environments above 60 °C or below -10 °C.

Q8 Is the robot dog waterproof? What should be done if it gets rained on or submerged in water?

This product is not waterproof. Do not submerge the unit. Do not clean it with high-pressure water jets or hoses; fire or explosion hazards may occur. If rain exposure or submersion happens while powered on, cut power immediately to stop current flow. If rain-soaked or submerged while powered off, remove the battery right away. Do not attempt to power on or charge the unit, as internal circuits may be damaged. Contact after-sales service as soon as possible.

Q9 The robot dog wobbles or assumes abnormal postures while walking. How to resolve this?

(1) This is often caused by non-zeroed joint motors: Manually place the robot dog limbs into the standard prone posture. Tap Joint Reset in the APP to zero joint motors, then test walking again.

(2) If the problem persists, motor joints may have been assembled incorrectly. Cross-check motor installation against the assembly manual and motor labels.

Q10 How do I download logs for after-sales support in case of unknown faults?

(1) Connect your mobile phone to the robot dog's Wi-Fi.

(2) Open the BAOBOT APP and go to the device page. Tap Connect, then enter Device Settings.

(3) Select System Log → Download Log.

(4) If tapping System Log opens a browser with no response: stay connected to the robot-dog Wi-Fi (BPX-xxxxxx). Open a browser and visit **http://10.21.40.1:8080/** or **http://10.21.20.1:8080/**, then proceed to download logs.

(5) Send the downloaded log file to official technical support for analysis.

Q11 What information can be viewed via the APP fault-detection function?

After the APP connects to the device, navigate to Device → Fault Detection. Real-time status is displayed for the four legs: Left-Front (LF), Right-Front (RF), Left-Rear (LR), Right-Rear (RR), the IMU sensor and the battery. You may also review historical fault records showing fault description, occurrence time and clearance time. Follow on-screen guidance for troubleshooting.

Q12 The robot dog suddenly lies prone during movement. Is this a fault?

This is usually thermal protection triggered by over-heated M-XX motors. Let the robot dog rest in prone position for a period before retrying. Check for sustained overloading (maximum permissible load: 5 kg), prolonged high-speed motion or extreme postures. Adjust operating intensity accordingly.

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